AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2013

Biology Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Questions 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

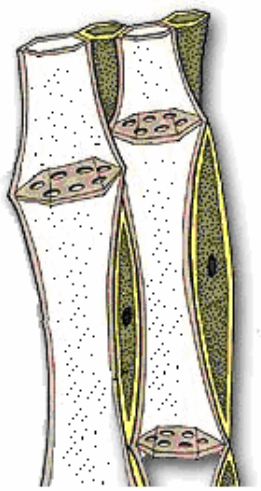
3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

a. The diagram shows a tissue found in plants. (2 Marks)



Mention any TWO special features which help this tissue to perform its function efficiently.

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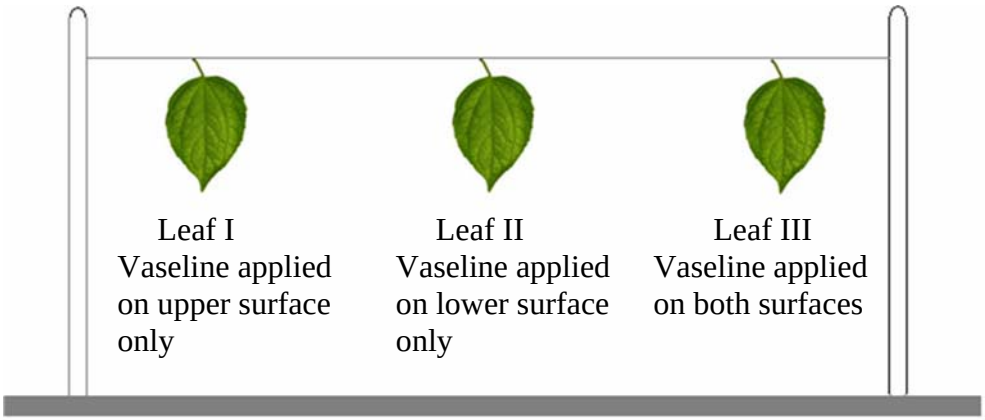
b. Describe the phenomenon of active transport on the basis of the following. (3 Marks)

	Active Transport
Movement of Molecules	
Requirement of Energy	
Example	

Q.2. (Total 4 Marks)

An experiment was conducted to find out the amount of water loss from two different surfaces of leaves.

The set up of the experiment is shown below.



The results of the experiment are as follows.

Leaf No.	Initial Mass/g	Final Mass/g	Difference in Mass/g
I	3.5	2.1	1.4
II	3.5	3.0	0.5
III	3.5	3.5	0.0

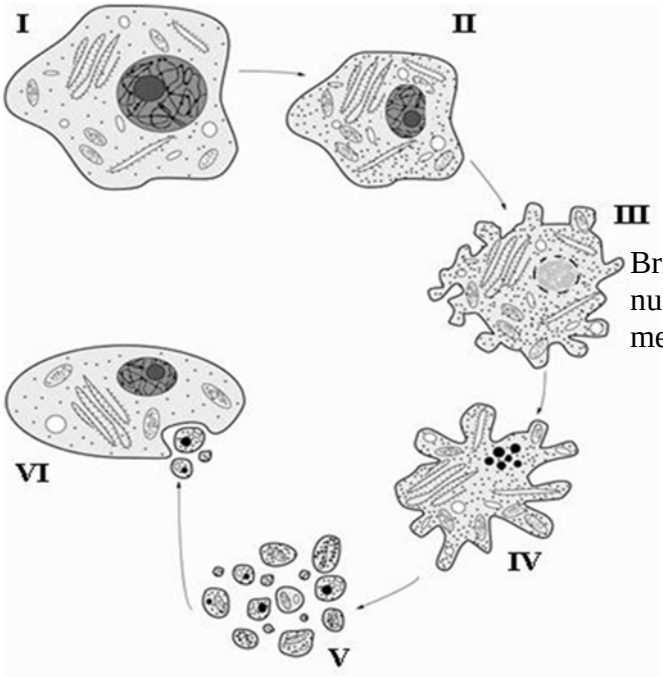
a. What is the purpose of leaf III in this experiment? (1 Mark)

b. Based on the given results, draw the conclusion for this experiment. (3 Marks)

Q.3. (Total 3 Marks)

The given diagram shows some events of apoptosis. Briefly describe what happens at stages II, IV and VI.

Break down of cytoskeleton



Stage II: _____

_____

Stage IV:_____

_____

Stage VI:_____

_____

Q.4. (Total 2 Marks)

What happens to the amount of enzyme at the end of a chemical reaction? Why does it happen?

Q.5. (Total 6 Marks)

a. Briefly describe the synthesis of NADPH in the light reaction of photosynthesis. (3 Marks)

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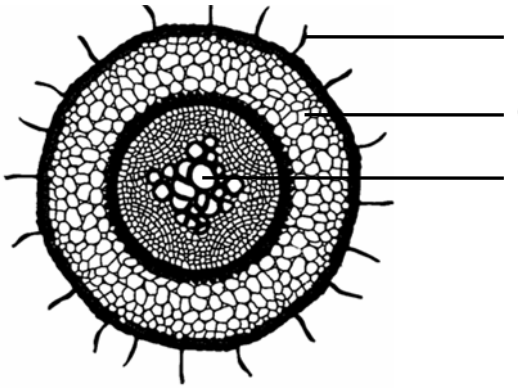
b. Complete the given table by providing required information about peptic ulcer. (3 Marks)

	Peptic Ulcer
Symptom	
Cause	
Treatment	

Q.6. (Total 5 Marks)

The diagram shows the internal structure of root.

a. Identify structures F, G and H. (3 Marks)



F _____

G _____

H _____

b. What is the role of structure F in plants? (2 Marks)

_____

_____

_____

_____

Q.7. (Total 7 Marks)

EITHER

a. What do you understand by the term ‘division of labour’? Describe cellular organization in a multicellular organism.

OR

b. Describe the significance of biodiversity with reference to the maintenance of the ecosystem and recycling of nutrients.

Q.8. (Total 8 Marks)

EITHER

a. Explain why the dietary requirements of a teenager are different from an old man.

(Your answer should include distinguishing food components of both individuals)

OR

b. Name and locate the valves present in heart and their associated structures. Explain the working of these valves to bring about normal blood circulation.

END OF PAPER

Please use this page for rough work